

Kinematic Scheme of Lathe Machine

Speed Gear Box

Head stock

Tool Post

Saddle

Chuck

Tail Stock

Feed rod

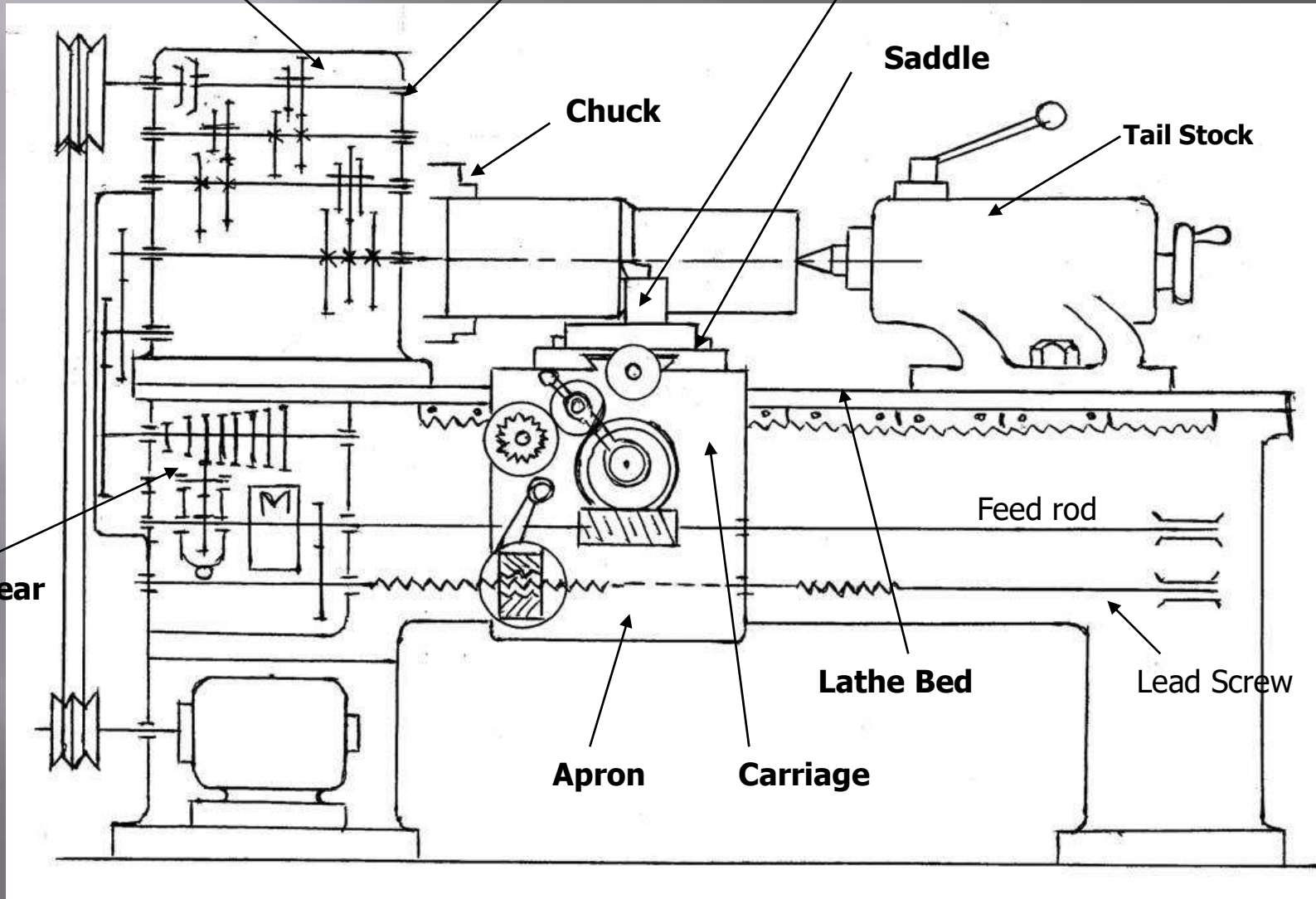
Lathe Bed

Lead Screw

Apron

Carriage

Feed Gear Box



Lathe Machine – Pulley driven (Light Duty)

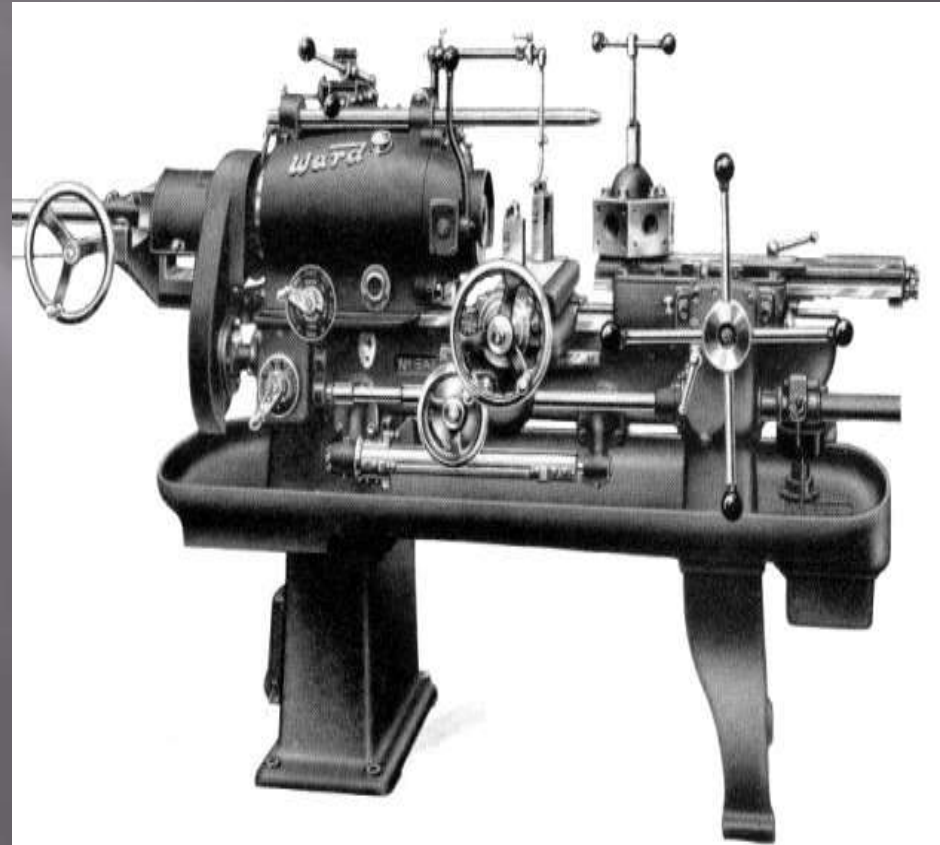


Lathe – All geared



- ▣ A capstan lathe or a turret lathe is a production lathe used to manufacture any number of identical pieces in the minimum time.
- ▣ These lathes are development of engine lathes.
- ▣ The capstan lathe was first developed in the United States of America by Pratt and Whitney sometimes in 1860.
- ▣ Special characteristics of a capstan or turret lathe enable it to perform a series of operations such as drilling, turning, boring, thread cutting, chamfering, cutting- off and many other operations in a regular sequence to produce a large number of identical pieces in a minimum time.

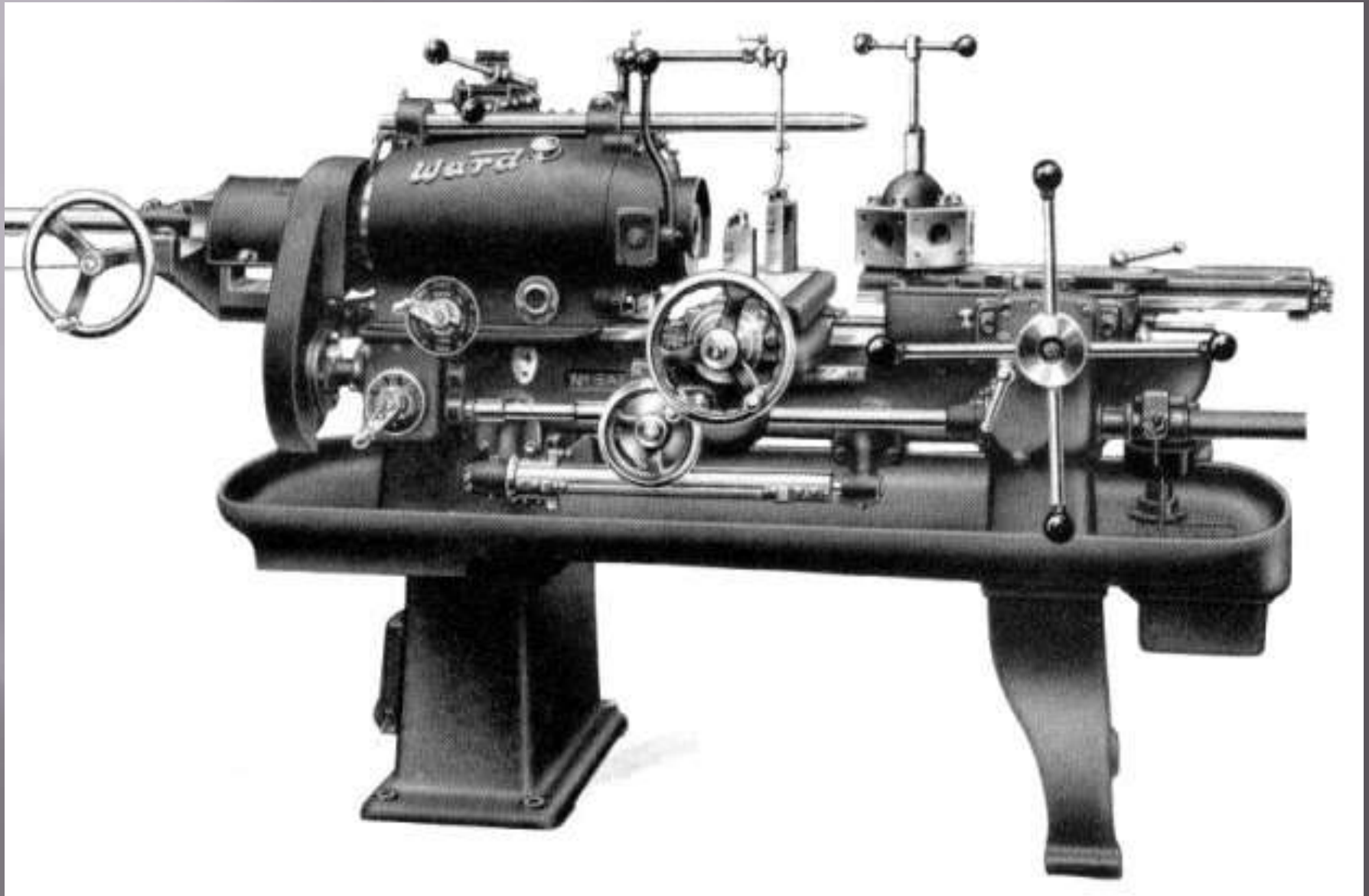
Difference Between Center lathe & Capstan, Turret Lathe



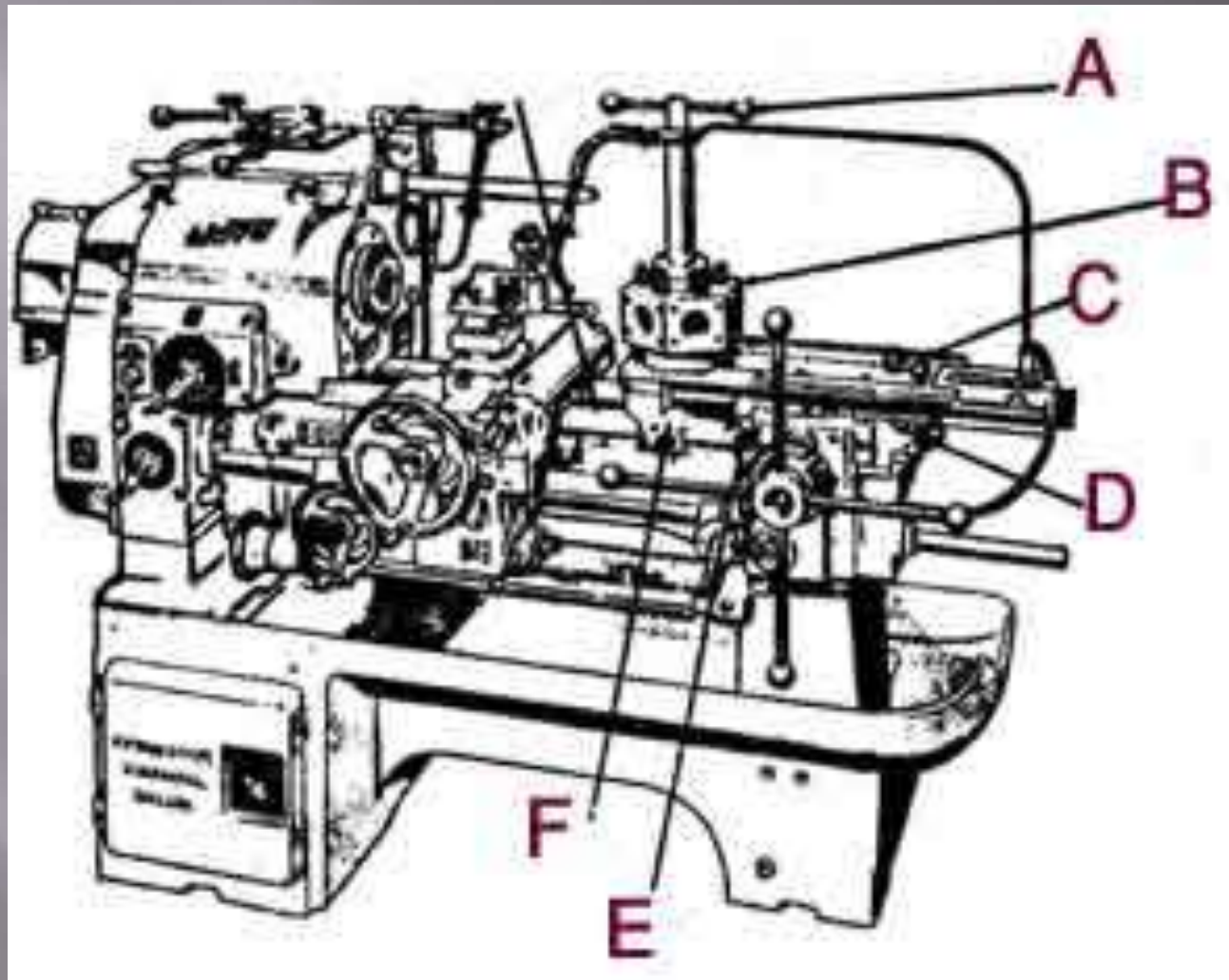
Centre Lathe	Capstan and turret Lathe
It is a manually operated lathe	It is a semi automatic lathe
It has only one tool post. so tool changing time is more	Front and rear tool posts are available. Tool changing time is less
It has tail stock	It has turret head instead of tail stock
Only one tool can be fitted in the tail stock	Six different tools can be fitted in the turret head.
Number of speeds is less	Number of speeds is more
Tool changing time is more	Tool changing time is less
The machine should be stopped for changing tool	Tool can be changed without stopping the machine
It is not suitable for mass production	It is suitable for mass production
No feed stops to control the tool	The tools are controlled by feed stops
The tool is centered manually after changing the tool	The tool is centered automatically
Only one operation is done at a time	More than one operation can be done at a time

Capstan lathe	Turret lathe
It is a light duty machine	It is a heavy duty machine
The turret head is mounted on the ram and the ram is mounted on the saddle.	The turret head is directly mounted on the saddle and the saddle slides over the bed ways
The saddle will not be moved during machining	The saddle is moved along with the turret head during machining
The lengthwise movement of turret is less	The lengthwise movement of turret is more
Short work pieces only can be machined.	Long work pieces can be machined
It is easy to move the turret head as it slides over the ram	It is difficult to move the turret head along with saddle
The turret head cannot be moved crosswise	The turret head can bbe moved crosswise in some turret lathes
As the construction of lathe is not rigid, heavy cut cannot be given	As the construction of lathe is rigid, heavy cut can be given
It is used for machining work pieces up to 60mm diameter	It is used for machining work pieces up to 200mm diameter
Collection of lathe bed lathe bed lathe bed	

Capstan Lathe







A – Turret Locking Handle, B – Turret, C - Capstan Slide,
E - Capstan Slide Block, D - Capstan Rest, F - Sliding Bridge

Turret Lathe



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Vertical Turret Lathe – CNC 16' dia



Vertical Turret Lathe – 88" Dia Table (2 Turret heads)



Vertical Turret Lathe – CNC 46" dia



Principal parts of capstan and turret lathes

- The turret has essentially the same parts as the engine lathe except the turret and the complex mechanism

Different parts of Capstan lathe are

- Head stock
- Cross-slide tool post
- Hexagonal turret
- Saddle for auxiliary slide
- Auxiliary slide
- Feed rod
- Saddle for cross-slide.

Different parts of turret lathe are:

- Head stock
- Cross-slide tool post
- Hexagonal turret
- Turret saddle
- Feed rod
- Saddle for cross-slide.

Bed:

- The bed is a long box like casting provided with accurate guide ways upon which are mounted the carriage and turret saddle.
- The bed is designed to ensure strength, rigidity and permanency of alignment under heavy duty service

Headstock:

- The headstock is a large casting located at the left hand end of the bed. The headstock of capstan and turret lathe may be of the following types.
 - 1.Step cone pulley driven headstock
 - 2.Direct electric motor driven headstock
 - 3.All geared headstock

CROSS-SLIDE&SADDLE:

- The larger capstan lathes and heavy duty turret lathes equipped with usually two designs of carriage
- 1.Conventional type carriage 2.Side hung type carriage
- **Conventional type carriage** bridges the gap between the front and rear bed way sand is equipped with four station type tool post at the front, and one rear tool post at the back of the cross-slide.
- **The side hung type carriage** is generally fitted with heavy duty turret lathes where the saddle rides on the top and bottom guide ways on the front of the lathe bed